

Ng Yu Hueng

Mobile: (+65) 81615806 • Email: yuhueng.business@gmail.com • LinkedIn: [linkedin.com/in/ngyuhueng](https://www.linkedin.com/in/ngyuhueng)
GitHub: github.com/yuhueng • Personal Website: ngyuhueng.com

EDUCATION

Singapore University of Technology and Design (SUTD)

Sep 2022 to May 2026

- Bachelor of Engineering (Computer Science and Design) — Tracks: Data Analytics, FinTech; Minor in AI
- GPA: 4.12 / 5.0 (Honours with Distinction)
- Relevant Courses:** Algorithms, Java Programming, Machine Learning, Artificial Intelligence, Database Systems, Natural Language Processing, User Interface and Implementation, Cloud Computing & IoT

TECHNICAL SKILLS

- Languages, Frameworks & Libraries:** Python • JavaScript • HTML • CSS • Ruby • Java • C • SQL • React • ExpressJS • FastAPI • Rails • Flask • TailwindCSS • Bootstrap • Streamlit • Pandas • NumPy • PyTorch • HuggingFace • YOLOv5
- Databases & Cloud/Tools:** MongoDB • Firebase • DynamoDB • AWS • Google Cloud • Databricks • Docker • Git • Vercel

WORK EXPERIENCE

Singtel Singapore, Application Dev & Ops Intern (Management Associate Program Intern)

May 2025 to Aug 2025

- Shipped an email summarization API in TypeScript (Azure Prompt Flow, OpenAI) that parsed MIME payloads and served concise summaries to internal teams via the enterprise Data Lake
- Built an Evaluation Metric API using Prompt Flow and Ragas to assess the effectiveness of existing AI prompts against metrics such as faithfulness and answer relevancy
- Developed test cases for Shirley Voicebot to validate functionality, user interactions, and conversation-flow edge cases
- Contributed to a business project focused on ideating strategies to improve GOMO's market attractiveness among the youths

pQCee Pte Ltd (Post-Quantum Cybersecurity Startup), Software Engineering Intern

Feb 2025 to May 2025

- Integrated post-quantum cryptographic APIs into existing solutions — FormSG, Outlook, and Gmail via a Chrome extension
- Developed demo web app (PWA) and backend APIs to showcase pQCee's solutions and wrote a [blog](#) documenting the work

PROJECTS

Capstone Project with ESGpedia ([Website](#)) | AWS, Databricks, Python, JavaScript

Sep 2025 to May 2026

- Built an ESG data extraction pipeline on Databricks using LLMs (Qwen-VL, Perplexity, OpenAI) to process sustainability reports from SGX-, Bursa Malaysia- and IDX-listed companies at ~97% field completeness and 86.1% extraction accuracy
- Engineered a data post-processing layer standardizing extracted ESG metrics into common units and a comparable schema
- Shipped a Flask/FastAPI backend with S3 integration, powering an extraction dashboard, anomaly detection on ESG metrics, and an AI chatbot for ESGpedia's sustainability intelligence platform

Natural Language Processing Project ([GitHub](#) | [Demo](#)) | Python, JavaScript, HuggingFace

Sep 2025 to Dec 2025

- Fine-tuned Qwen3-4B into a Singlish base model using 4-bit QLoRA on a self-constructed dataset, with perplexity of 16.26
- Trained 4 persona adapters evaluated via perplexity and human evaluation. Deployed via HuggingFace ZeroGPU Spaces
- Built a dating chat simulator (React, FastAPI) with LionGuard content moderation, deployed on Vercel and Render

FabWatcher, Cloud & IoT Project | AWS, DynamoDB, C, Python, JavaScript

Sep 2025 to Dec 2025

- Designed a 3D printer management system with booking-triggered workflows via AWS EventBridge, Lambda, and MQTT
- Implemented real-time print failure detection via YOLO for anomaly classification, streaming telemetry to DynamoDB
- Deployed on AWS with a containerized EC2 backend and a static S3 frontend, with a live admin dashboard

Portfolio Website ([GitHub](#) | [Website](#)) | JavaScript, MongoDB, React, ExpressJS, NodeJS

Jul 2024 to Jul 2024

- Built and deployed a MERN-stack portfolio on Vercel and DigitalOcean, with form validation and automated email handling

DBS DocCheck Web App | Ruby on Rails, JavaScript, HTML, CSS, Google Cloud

May 2024 to Aug 2024

- Built a Ruby on Rails app on Google Cloud streamlining DBS's credit card application, delivered in bi-weekly Agile sprints
- Integrated Vertex AI for document verification (passports, employment passes) and for a customer support chatbot
- Employed Document AI to extract fields (name, gender, etc.) from identity documents and auto-populate application forms

LeftoverChef Android App ([GitHub](#) | [Demo](#)) | Java, XML, Firebase, YOLOv5, TensorFlow, OpenAI

Jan 2023 to Apr 2024

- Engineered an Android app to mitigate wastage, using OpenAI for recipe generation, and Firebase for real-time data storage
- Awarded Honorable Mention and a \$1,000 prize at the Singtel Information Systems and Programming Merit Award

CERTIFICATIONS

IBM RAG and Agentic AI Professional Certificate (Coursera)

Jul 2026 to Aug 2026

- Built agentic systems with LangChain and LangGraph: tool calling, memory, and self-improving agents (Reflection, Reflexion, ReAct), plus multi-agent orchestration with CrewAI, AutoGen, BeeAI and MCP
- Implemented RAG pipelines with advanced retrievers and vector databases (FAISS, ChromaDB)